

Chaitanya Mamatha Ananda

Compiler Optimization Engineer | C++ / LLVM | Ph.D. Candidate

Riverside, CA | 951-907-8519 | cmama002@ucr.edu | [LinkedIn](#) | [Google Scholar](#)

Summary

Computer Science Ph.D. candidate specializing in LLVM-based compiler optimization, with a research focus on improving data and code locality. Experience in profile-guided optimization, post-link binary optimization, and AI-driven code-layout optimization.

Technical Skills

Programming: C++, C, Python, NumPy; Linux, Git

Compilers: LLVM, BOLT, Propeller, profile-guided optimization (PGO/FDO), post-link optimization, interprocedural optimization, basic block reordering and deduplication, code layout, heap locality

Performance: x86, Arm (AArch64), perf, hardware performance counters, profiling, benchmarking

ML for Compiler Optimization: AlphaEvolve, Vizier, MLGO, Evolution Strategies, heuristic search

Experience

Google | Student Researcher

Sep 2025 – May 2026

Compiler optimization team

Sunnyvale, CA

- Developed **AI-PROPELLER**, extending Google's **Propeller** with AI-driven interprocedural code-layout optimization for warehouse-scale binaries; used **AlphaEvolve** and hardware counters to guide heuristic search.
- Reduced execution time by **1.6% on LLVM Clang** and **0.23% on a production Search workload**.

University of California, Riverside | Ph.D. Researcher

Sep 2022 – Present

Compiler optimization research; Advisor: Prof. Rajiv Gupta

Riverside, CA

- Developed **PreFix**, a profile-guided heap-layout optimization that groups hot objects to improve data locality; reduced execution time by **21.7% on average**.
- Developed **DeduBB**, a post-link basic block deduplication technique implemented in **BOLT** and **Propeller** for x86 and AArch64; reduced binary size by **up to 25.8%**.
- Handled stack-manipulating code across modules; used profiles to deduplicate cold blocks and preserve performance.

Indian Institute of Science | Undergraduate Researcher

Aug 2019 – May 2022

Parallel programming and scientific data analysis

Bengaluru, India

- Built a parallel programming model for PDE solvers using **Regent/Legion**; developed a scientific-data anomaly detector using statistical and deep-learning methods.

Robert Bosch Engineering and Business Solutions | Project Trainee

Spring 2021

Developed scripts to synchronize video and radar data collected during automated test driving.

Publications

PreFix: Optimizing the Performance of Heap-Intensive Applications.

C. Mamatha Ananda, R. Gupta, S. Tallam, H. Shen, and X. D. Li.

CGO 2025 | [Paper](#)

DeduBB: Binary Code Size Reduction via Post-Link Basic Block Deduplication.

C. Mamatha Ananda, M. Afarin, R. Gupta, S. Tallam, H. Shen, and X. D. Li.

LCTES 2026 | [Paper](#)

AI-PROPELLER: Warehouse-Scale Interprocedural Code Layout Optimization with AlphaEvolve.

C. Mamatha Ananda, R. Gupta, M. Trofin, A. Grossman, S. Tallam, X. D. Li, and A. Yazdanbakhsh.

MLArchSys 2026; CAIS AID-Wild 2026 (workshops) | [Paper](#) ([arXiv](#))

Patent

Generate Better Code Layout Heuristics | Co-inventor, Google | Patent filed June 2026

Education

Ph.D. in Computer Science, University of California, Riverside

Sep 2022 – Present

B.E. in Computer Science and Engineering, Bangalore Institute of Technology, India

2021

Awards & Teaching

Fellowship Dean's Distinguished Fellowship, UC Riverside

2022

Teaching Assistant Compiler Design and Construction, UC Riverside